

An Analysis of Kinect-Based Human Fall Detection System

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Abstract— Human fall detection system has become one of the most important things especially for indoor environment application. This system has been used in respective areas of elderly care and at child care houses. It helps to detect any human fall and will alert the caretaker about the accident. Kinect sensor can be used to perform the detection due to its capability in scanning and tracking human as well as its affordability. One of the widely used algorithm in human fall detection using Kinect is the skeleton-based method where it works by calculating the distances of every joint with the floor-plane. The joints are detected using the skeleton space coordinate system. When the floor-plane is not visible and the Y-coordinate is less than the given value, a fall is detected. Due to its widely usage, there is a need to study its performance to know the best condition that this algorithm could offer. Performance of selected parameters were observed through a few experiments conducted using Visual Studio as the interface. In this work, a mobile-based Kinect is used due to its mobility and better future implementation for indoor navigation. The best parameter can be identified quantitatively in order to choose the ideal scene that can be used to detect human fall detection using this skeleton-based method. Among the parameters are the distance of the human to the Kinect, the light intensity, the time to track human and the speed of fall. It can be concluded that the most ideal conditions would be at a distance of 3 meters to 3.5 meters with lightings of 1007 lux and of 2 persons at the scene. These conditions can be helpful for others when considering to use the algorithm for human fall detection using Kinect.

Keywords— *human fall detection, skeleton-based human detection, Kinect-based*

I. INTRODUCTION

Fall always happen even when human in conscious situation where they can trip and fall while walking, fall from bed when sleeping, or instable when walking. According to the official portal MyHEALTH of Ministry of Health, Malaysia [1], over third of the elderly living in homes died annually due to fall. At the old folks homes and child care homes, there are many cases where the elderly and child have been found falling without their caretaker realize about the situation. This situation can cause a serious harm to the elder and the child if they fall on a sharp things or dangerously injuring their head. To avoid this situation, fall detection system has been introduced to reduce risks of fall. This system helps to alert the care taker if the accident happened. Various types of approach has been made to handle this situation. One of the approaches is by wearing a sensor that can be used to detect acceleration of the falls [2]. However, if the elderly forgot to wear the sensor, no falls can be detected even when the incident happened.

Kinect is one of the sensor that able to be utilised for human fall detection due to its advantages in providing depth images. Due to this, Kinect able to recognize human faces, capture full-body 3D motion, and capable to recognize voice based on the collection of sensors in it [3]. It can also be used to assist robots in avoiding obstacles and follow human [4]. By using Kinect, robot able to interpret its surrounding and use it as its own eyes to detect the obstacles by avoiding or following it. According to [5], it is also one of the most developed devices in terms of image acquisition. This 3D camera enables natural human-robotic interaction / interface (HRI) to be developed as it generates a depth map. Such HRI systems can recognize various gestures performed by a human operator, making the interaction process easier. This allows the robots to be easily and naturally controlled [6].

Due to the above advantages, this project will utilise a mobile-based Kinect platform for human fall detection. One of the ways to detect human fall is through skeleton-based method. Although widely used, studies on the best parameters for human fall detection has not being performed, and this is crucial in ensuring the effectiveness of the adapted method. Suitable parameters are selected based on the workability of the sensor used, i.e. light intensity, distance of human to the sensor, time taken for human detection and number of persons detected.

II. METHODOLOGY

A. Kinect Sensor

Kinect sensor is used in this project based on its features and ability to detect and track human with affordable cost. Kinect provides RGB camera, depth sensor, multi-array mic and motorized tilt. It uses Windows SDK and for this project, Visual Studio software will be used. Due to the advantages offered by mobile robots in indoor applications, the Kinect sensor will be placed on a mobile platform.

B. Mobile Platform

The mobile platform used in this project can be seen in Fig. 1. The platform can be controlled via mobile application through voice or the arrow pad as in the application for maneuverability. More information on the mobile platform can be obtained from [7].



Fig. 1. The mobile-based Kinect used in this work

C. Visual Studio Software

Visual Studio software is one of the Microsoft integrated development which is functions to develop website, computer program, web applications, web service and mobile applications. This software can be used to develop a program, analyzed, debug, test, collaborate and deploy. The IntelliSense code suggestion can help to minimize the error while building the program.

This software provides a compiler that can be used with any kind of interface and multiples programming language that can be choose from. For the human fall detection system, the C# language is chosen as it is one of the fastest compiler in C programming language [8]. Fig. 2 shows the front panel used for this project which is built by using Visual Studio. It will display the results according to the human joint to the floor-plane, average velocity and the system status.

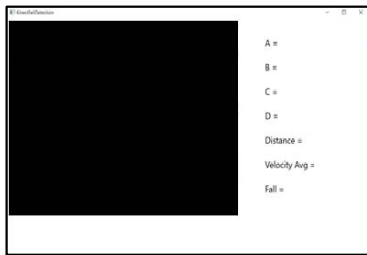


Fig. 2. Front panel of the human detection system

D. Human Tracking Method

Existing human tracking methods have various way to detect the human and have been used in many kinds of applications especially for security systems, gaming and human following robot. Human tracking methods that will be used in this project is the human skeleton algorithm. According to previous researchers, this method is one of the best methods to track human using Kinect sensor [9].

The skeleton algorithm used in this project is by defining the distance of the tracked skeleton joint to the floor plane in detecting a fall. The human joints can be seen as in Fig. 3. The equation used to calculate the distance of the joints to the floor plane can be referred as in (1). The algorithm states that human can be detected as falling if the distance less than 0.3 meter from the floor plane [9].

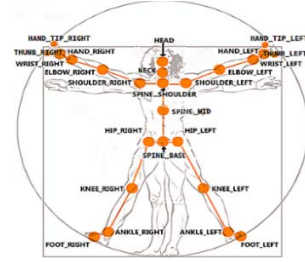


Fig. 3. The human joints point

$$d = \frac{Ax_0 + By_0 + Cz_0 + D}{\sqrt{A^2 + B^2 + C^2}} \quad (1)$$

The velocity for every joint can also be calculated for detecting fall and is acquired from (2). The velocity must be known in order to know the average velocity of the joints.

$$v = \frac{d}{t} \times 1000 \quad (2)$$

Then, the average velocity can be calculated as shown in (3). This equation will determine the threshold value that will be applied to the algorithm and to determine the fall status.

$$v_{avg} = \frac{\sum v}{n} \quad (3)$$

To detect a fall, the thresholding method is used. It works by checking the average velocity when the head and ankle is on the floor and the value of head, shoulder centre, hip centre, ankle left and ankle right are smaller; otherwise a safe situation is considered.

III. RESULTS AND DISCUSSION

The project is conducted on a normal indoor environment (laboratory) equipped with lightings. Table I shows the number of fluorescent lamp used and the light intensity recorded by the lux meter at the environment where the experiments were conducted.

TABLE I. LUX METER READING ACCORDING TO NUMBER OF FLUORESCENT LAMPS

No. of Fluorescent Lamp Turned On	Light Intensity (lux)
16	1007
12	930
8	489
4	101

To study the effect of lighting and distance with respect to human detection, the time taken to detect human joint for different light intensity and distance of the human to the Kinect sensor is measured and tabulated in Table II. To minimize the error during data collection, the same person is used during this data collection. Distance chosen here are based on the specification of Kinect in human detection, which is between 2 to 4 meters.

TABLE II. TIME TAKEN TO DETECT HUMAN FOR DIFFERENT LIGHT INTENSITY AND DISTANCE

Distance (m)	Light Intensity (lux)				Average
	1007	930	489	101	
	Time (s)				
2.0	0.67	0.66	0.72	0.63	0.67
2.5	0.53	0.60	0.63	0.69	0.61
3.0	0.57	0.50	0.53	0.66	0.57
3.5	0.56	0.56	0.72	0.47	0.57
4.0	2.04	4.79	1.73	3.23	2.95
Average	0.87	1.42	0.87	1.14	

From the data in Table II, it can be seen that the most ideal condition to detect the human joint is at the distance of 3 meters and 3.5 meters with light intensity of 1007 and 489 lux. To further study on these effects, analysis of variance (ANOVA) is used. Assuming:

H_0 : There is no effect of lighting intensity on the distance to the sensor

H_1 : There is an effect of lighting intensity on the distance to the sensor

Test statistic, F, and critical value, F_{cv} can be determined:

$$F_{test} = 10.631$$

At $\alpha = 0.05$, $k=9$, $N=36$ and using statistical table for F-distribution, $F_{0.05,8,24} = 2.355081$.

Since $F_{test} = 10.631 > F_{cv} = 2.355081$, thus we can reject H_0 . Hence, there is sufficient evidence to support the claim that there is an effect of lighting with the distance to the sensor as can be seen from Fig. 4.

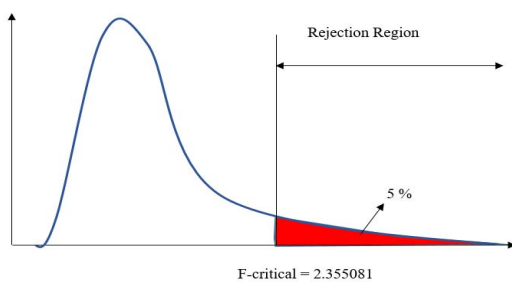


Fig. 4. The F-Distribution rejection region of treatment effect

For the next experiments, the lighting condition which produces the best results is chosen, which is 1007 lux. Although both 1007 and 498 lux conditions give the same average results of detection with 0.87 s, 1007 lux is chosen as Kinect is a light-based sensor, thus working in a brighter condition is preferred. Algorithm used for the human fall detection system is tested in this condition and the system accuracy is recorded. This project used the thresholding method to detect the human fall in the algorithm. The threshold value and average acceleration to detect the system fall status is tabulated as in Table III.

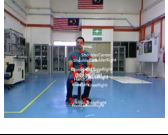
According to Table III, the average velocity of the human joint point will determine the threshold value to be applied for the algorithm and the output of the fall system status. If the average velocity of the human joint points more than 0 m/s, then there will be no threshold value and the system detects no fall at the surrounding. However, if the average velocity of the human joint points lower than 0 m/s, then the threshold value is -0.1, thus the system status will declare a fall.

TABLE III. AVERAGE VELOCITY TO DETECT FALL STATUS

Average Velocity (m/s)	Threshold	System Fall Status
≥ 0	-	False
< 0	-0.1	True

The scenario of the system performance is described in Table IV. The system able to detect fall when there were fall happened at the range of the Kinect sensor. The front panel of the system will display and alert the user with red warning when there is a fall detected at the surrounding.

TABLE IV. DIFFERENT POSTURE OF THE TARGET PERSON

Posture	Descriptions	System Status
	- One person stands still - Walking / running around	Fall=False
	- Two persons stand still - Walking / running around	Fall=False
	One person is sitting on a chair in a fix place	Fall=False
	- One person falls in a fix place - Fall in sitting posture	Fall=True
	- One person falls in a fix place - Fall on the ground with lying posture	Fall=True
	- Two persons stand in front of the sensor - Walking around - One person slips and fall while the other still standing	Fall=True

The parameters used in this project are the average velocity of skeleton joint and its relationship with threshold value in the algorithm to find the fall detection system status. The result of average velocity to detect fall for different distance was recorded according to Table V. From the table, when the distance of the human is nearby the limit of Kinect, it fails to detect any fall. The Kinect sensor does not perform

well when the human is not in the ideal distance range and this can cause error (NIL).

TABLE V. AVERAGE VELOCITY TO DETECT FALL FOR DIFFERENT DISTANCE

Distance (m)	Average Velocity (m/s)	
	Fall=False	Fall=True
2.0	0.0395874	NIL
2.5	0.0107593 (≈ 0)	-0.1044836
3.0	-0.0038227 (≈ 0)	-0.2967304
3.5	-0.0013795 (≈ 0)	-0.3097800
4.0	0.0005469	NIL

The human fall detection result is summarized as in Table VI. If a human falling at distance 2 meters and 4 meters, there are no falls detected even after 3 times of falling down trial. However, distance of 3 meters and 3.5 meters are the best performance shown by the Kinect sensor as it able to detect all the falls that happened.

TABLE VI. FALL DETECTION RESULTS

Distance (m)	No. of Simulated Falls Scenario	No. of Detected Falls Scenario	No. of Undetected Falls Scenario	Accuracy (%)
2.0	3	0	3	0.00
2.5	3	1	2	33.33
3.0	3	3	0	100.00
3.5	3	3	0	100.00
4.0	3	0	3	0.00

Fig. 5 and 6 shows the front panel display when there are human in front of the sensor. Alphabet “A”, “B”, “C” and “D” shows the parameters applied to (1) to detect the distance of the human joint points to the floor-plane. “Velocity Avg” is the average velocity of the human joint points while “Fall =” will show the result of the fall detection either “True” or “False”.



Fig. 5. Front display panel for fall detected



Fig. 6. Front display panel for no fall detected

As a summary, this project analyzes the performance of the chosen algorithm in tracking human and detecting human fall quantitatively to be used as a reference for future project which utilize the same algorithm. The system can track the human joint the fastest when the light intensity is 1007 lux with the distance of 3 meters and 3.5 meters. The tested algorithm has successfully detected human falls at 100% accuracy at these parameters and able display the result accurately. It can be concluded that the chosen Kinect-based algorithm able to detect any human fall at its surrounding within the parameters and alert the user about the incident.

IV. CONCLUSION

This project is to study on the performance of existing Kinect-based human fall detection system. It is helpful as it can be used at the elderly homes or child nursing homes in detecting human fall. The chosen algorithm detects human fall using skeleton-based and thresholding algorithm. This system used Kinect sensor to track and detect for human fall. The quantitative measurement is done to identify the ideal conditions in detecting the human fall. Several parameters have been chosen to test the algorithm in detecting human condition for different posture. The system status will display the situation of the human whether is it a fall or vice versa.

The algorithm is programmed using C# programming language in Visual Studio software and interfaced with Window SDK as the intermediate software for Kinect sensor. These software work perfectly together and provide less communication error between the algorithm and the sensor. The system able to show accurate results within the ideal conditions.

Based on the gathered data, the algorithm has successfully displayed true status for every different posture and situation of the human at ideal conditions of 3 meters and 3.5 meters from the Kinect at light intensity of 1007 lux. Analysis of Variance (ANOVA) is used as the statistical analysis to prove the relationship of the parameters used is strong enough to be evaluated.

As for the future improvement and recommendation, a dynamic environment can be considered to measure the performance of the chosen algorithm in detecting human fall.

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